Emil Kasper

Vienna and Zurich kaspere@ethz.ch emil-kasper

Experience

Franka Robotics, Machine Learning Engineer (Intern) — Munich, Germany
- Designing and running machine-learning training environments.
- Developing and integrating software systems, and prototyping in support of applied AI development.

Oct 2026 – present
1 month

TU Wien Robotics Club, Machine Learning Engineer — Vienna, Austria
- Implementing vision-language-action models (VLAs) and world models for robotics tasks.
- Developing vision algorithms for a chess robot to autonomously move the pieces.

Apr 2026 – present
7 months

IVIA Lab, ETH Zurich, Research Fellow (ETH SSRF) — Zurich, Switzerland
Supervised by Prof. Menna El-Assady.
- Selected as one of 20 students from over 4,000 applicants for the ETH Student Summer Research Fellowship.
- Working on human-agent and agent-agent collaboration in agentic AI systems.
- Co-authored a paper submitted to ICLR 2027.

July 2026 – Aug 2026
2 months

Siemens, Software Engineer — Vienna, Austria
- Contributing to a test simulation framework for the European Train Control System (ETCS).
- Implementing the autonomous train operation (ATO) mode in C# for test simulations.

July 2025 – June 2026
1 year

Institute of Visual Computing and Human-Centered Technology, TU Wien, Teaching Assistant — Vienna, Austria
- Guiding students through the lifecycle of user-centered product development.
- Providing feedback and grading for 45 students.

Mar 2025 – June 2026
1 year 4 months

Machine Learning Research Unit, TU Wien, Undergraduate Researcher — Vienna, Austria
Supervised by Prof. Clemens Heitzinger.
- Investigating the convergence and stability of Q-Learning and Speedy Q-Learning.
- Experimenting with PPO for robotic hand manipulation.

Aug 2025 – Apr 2026
9 months

TU Wien Space Team, Frontend Developer — Vienna, Austria
- Contributing to the "Lamarr" project at the 2025 European Rocketry Challenge (EuRoC).
- Building frontend tooling for rocket system control.
- Watch the successful launch at EuRoC 2025.

Mar 2025 – Oct 2025
8 months

Education

ETH Zurich, MSc in Computer Science — Zurich, Switzerland
- Minor in Machine Intelligence.

Sept 2026 – May 2028

TU Wien (Vienna University of Technology), BSc (Hons) in Computer Science — Vienna, Austria
- Four-year honors degree with a minor in machine learning.
- Graduated with distinction.
- Received merit-based scholarships.

Oct 2022 – May 2026

Research

Asking Your Advisor: Learning When to Consult Stronger Models in Agentic Systems
Submitted to ICLR 2027. We train a meta-policy that learns when to consult a stronger advisor model and when to continue with a cheaper one, reaching a better success-cost Pareto frontier than standard advisor-executor setups. Preprint on arXiv.

Engagement

European Student Robotics Association (ESRA), Executive Board Member, Partnerships and Events — Sept 2026 – present

- Leading partnerships and events for ESRA, a European network for student robotics.

European Student Assembly 2026, EUC Voices, Delegate — Strasbourg, France, Jan 2026 – Apr 2026

- Representing the EULiST alliance at the European Student Assembly 2026.
- Debating key issues and advocating for policy changes at the European level.
- Writing policy recommendations on empowering inclusive digital literacy and building resilience.

Skills

Programming
Python, Java, C, C++, C#, Bash, Linux

Machine Learning
Model development and evaluation

Reinforcement Learning
Algorithms, use cases, and applications

Computer Vision
Object detection and segmentation, visual perception for robotics

Software Engineering
Git, Docker, Maven, REST APIs

Human-AI collaboration
shared-initiative interaction, explainable AI, human-in-the-loop

Generative AI
Prompting, API-based integration, prototype development

Languages

German
Native

English
Fluent

Russian
Fluent